EE2016-B- Fall 2021

數位系統導論

Fundamentals of Logic Design-Chapter 3: Boolean Algebra (II)

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(p66)

3.1 Multiplying Out and Factoring Expressions (1)

- **The SOP (Sum-of-products) expressions can be obtained by multiplying out, using 2 distributive laws:**
  - The distributive law:

$$X(Y+Z) = XY + XZ \qquad (3-1)$$

  - The 2nd distributive law:

$$(X+Y)(X+Z) = X + YZ \qquad (3-2)$$

(p66) 3.1 Multiplying Out and Factoring Expressions (2)

- **The following theorem is very useful for factoring and multiplying out:**

$$(X+Y)(X'+Z) = \cancel{XX'} + XZ + X'Y + YZ$$
$$= XZ + X'Y + \cancel{YZ} = XZ + X'Y \quad (3-3)$$

(using Consensus)

$$(X+Y)(X'+Z) = XZ + X'Y \quad (3-3)$$

(p67) 3.1 Multiplying Out and Factoring Expressions (3)

- **Example 1:** $\boxed{(Q+AB')(C'D+Q')}$

$$(Q+AB')(C'D+Q')=QC'D+Q'AB'$$

If $(Q+AB')(C'D+Q')=QC'D+\cancel{QQ'}+\boxed{AB'C'D}+AB'Q'$

Very hard to eliminate this term!

(p67)

3.1 Multiplying Out and Factoring Expressions (4)

- **Example 2:**

$$\boxed{(A+B+C')(A+B+D)(A+B+E)(A+D'+E)(A'+C)}$$

$$(A+B+C')(A+B+D)(A+B+E)(A+D'+E)(A'+C)$$

$$=(A+B+C'D)(A+B+E)[AC+A'(D'+E)]$$

$$=(A+B+C'DE)(AC+A'D'+A'E)$$

$$=AC+\cancel{ABC}+A'BD'+A'BE+A'C'DE$$

AC(1+B)=AC →*Absorption*

(p67)

3.1 Multiplying Out and Factoring Expressions (5)

- **Example 3:**

$$\boxed{AC + A'BD' + A'BE + A'C'DE}$$

$$= \underbrace{AC}_{XZ} + \underbrace{A'}_{X'}(\underbrace{BD' + BE + C'DE}_{Y})$$

$$\because XZ + X'Y = (X + Y)(X' + Z)$$

$$= (A + BD' + BE + C'DE)(A' + C)$$

$$= [\underbrace{A + C'\mathrm{DE}}_{X} + \underbrace{B}_{Y}(\underbrace{D' + E}_{Z})](A' + C)$$

$$\because X + YZ = (X + Y)(X + Z)$$

$$= (A + B + C'DE)(A + \cancel{C'DE} + D' + E)(A' + C)$$

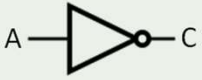
$$= (A + B + C')(A + B + D)(A + B + E)(A + D' + E)(A' + C) \quad (3\text{-}5)$$

(The same to (3-4))

$$\because C'\mathrm{DE} + \mathrm{E} = (\mathrm{C'D} + 1)\mathrm{E} = \mathrm{E}$$

(補充)

3.2 Basic Logic Gates and XOR Gate (1)

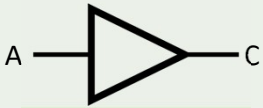
Type	Symbol	Logic Function	Truth Table	Explanation
AND (及)		$C = A \cdot B$	A B $A \cdot B$ 0 0 0 0 1 0 1 0 0 1 1 1	所有輸入為1時，輸出是1。 只要一輸入是0，則輸出是0。
OR (或)		$C = A + B$	A B $A + B$ 0 0 0 0 1 1 1 0 1 1 1 1	所有輸入是0，輸出才是0。 只要一輸入是1，則輸出是1。
NOT (非)		$C = \bar{A}$	A $C = \bar{A}$ 0 1 0 1 1 0 1 0	輸入與輸出的0/1狀態逆轉。
NAND (非及)		$C = \overline{A \cdot B}$	A B $\overline{A \cdot B}$ 0 0 1 0 1 1 1 0 1 1 1 0	所有輸入為1時，輸出為0。 只要一輸入是0，則輸出是1。

(補充)

3.2 Basic Logic Gates and XOR Gate (2)

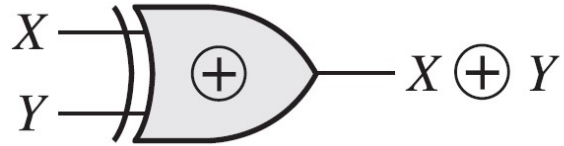
Type	Symbol	Logic Function	Truth Table	Explanation
NOR (非或)		$C = \overline{A + B}$	A B $A+B$ 0 0 1 0 1 0 1 0 0 1 1 0	所有輸入為0，輸出為1。 有一輸入為1則輸出為0。
XOR (互斥或)		$C = A \oplus B$	A B $A \oplus B$ 0 0 0 0 1 1 1 0 1 1 1 0	只能有其中一個輸入為1，輸出為1；否則輸出為0。
XNOR (反互斥或)		$C = \overline{A \oplus B}$	A B $\overline{A \oplus B}$ 0 0 1 0 1 0 1 0 0 1 1 1	只能有其中一個輸入為1，輸出為0；否則輸出為1。

(補充) 3.2 Basic Logic Gates and XOR Gate (3)

Type	Symbol	Logic Function	Truth Table	Explanation
BUF (同/是)		$C = A$	A / $BUF\ A$ 0 / 0 1 / 1	輸出一個與輸入相同的高低狀態。
IMPLY (蘊/含)		$C = A \rightarrow B$	A / B / $A \rightarrow B$ 0 / 0 / 1 0 / 1 / 1 1 / 0 / 0 1 / 1 / 1	如果第一輸入為低時，輸出高，否則輸出與第二輸入相同的高低狀態。
NIMPLY (非蘊/非含)		$C = \overline{A \rightarrow B}$	A / B / $\overline{A \rightarrow B}$ 0 / 0 / 0 0 / 1 / 0 1 / 0 / 1 1 / 1 / 0	如果第一輸入為低時，輸出低，否則輸出與第二輸入相反的高低狀態。

3.2 Basic Logic Gates and XOR Gate (4)

- **Truth Table of $X \oplus Y$:**



X	Y	X⊕Y	X'	Y'	X'Y	XY'	X'Y+XY'
0	0	0	1	1	0	0	0
0	1	1	1	0	1	0	1
1	0	1	0	1	0	1	1
1	1	0	0	0	0	0	0

$$\boxed{X \oplus Y = X'Y + XY' \ (3-6)}$$

(p69)

3.2 Basic Logic Gates and XOR Gate (5)

- **The following theorems apply to XOR:**

$$X \oplus 0 = X \quad \text{(BUF)} \tag{3-8}$$

$$X \oplus 1 = X' \quad \text{(Invertor)} \tag{3-9}$$

$$X \oplus X = 0 \quad \text{(相同為0)} \tag{3-10}$$

$$X \oplus X' = 1 \quad \text{(相異為1)} \tag{3-11}$$

$$X \oplus Y = Y \oplus X \text{ (交換律)} \tag{3-12}$$

$$(X \oplus Y) \oplus Z = X \oplus (Y \oplus Z) = X \oplus Y \oplus Z \text{ (結合律)} \tag{3-13}$$

$$X(Y \oplus Z) = XY \oplus XZ \text{ (分配律)} \tag{3-14}$$

$$\boxed{(X \oplus Y)' = X \oplus Y' = X' \oplus Y = XY + X'Y'} \tag{3-15}$$

(p70)

3.2 Basic Logic Gates and XOR Gate (6)

- **Proof of the distributive law:**

$$XY \oplus XZ = XY(XZ)' + (XY)'XZ$$

$$= XY(X' + Z') + (X' + Y')XZ$$

$$= \cancel{XYX'} + XYZ' + \cancel{X'XZ} + XY'Z$$

$$= X(YZ' + Y'Z) = X(Y \oplus Z)$$

3.2 XOR Gate and Equivalence Operations (7)(p69)

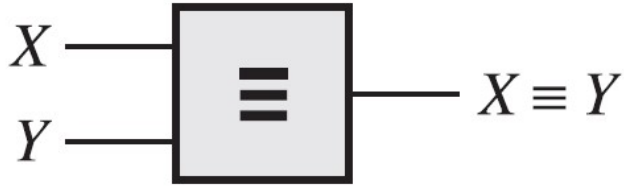
- **The equivalence operation (≡) is defined by:**

$$\begin{aligned}(0 \equiv 0) &= 1\\(0 \equiv 1) &= 0\\(1 \equiv 0) &= 0\\(1 \equiv 1) &= 1\end{aligned} \qquad (3\text{-}16)$$

3.2 XOR Gate and Equivalence Operations (8)(p69)

- **Truth Table of X ≡ Y:**

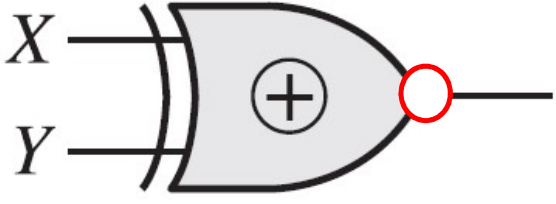
X	Y	$X \equiv Y$
0	0	1
0	1	0
1	0	0
1	1	1



(X ≡ Y)=1 iff X=Y=1

3.2 XOR Gate and Equivalence Operations (9)(p70)

- **An alternate symbol for the equivalence gate:**



$$(X \oplus Y)' = (X \equiv Y)$$

The equivalence gate is called an X-NOR gate

3.2 XOR Gate and Equivalence Operations (10)

(p70)

- **Example**

$$\boxed{A' \oplus B \oplus C} = [A'B' + (A')'B] \oplus C$$

$$= (A'B' + AB)C' + (A'B' + AB)'C$$

$$= (A'B' + AB)C' + (\cancel{AA'} + A'B + AB' + \cancel{BB'})C$$

$$= (A'B' + AB)C' + (A'B + AB')C$$

$$= A'B'C' + ABC' + A'BC + AB'C$$

3.3 The Consensus Theorem (1) (p70)

- **The consensus theorem (重合定理)** is useful in simplifying Boolean expressions. Given an expression of the form **XY+X'Z+YZ**, the term YZ is redundant and can be eliminated to form the equivalent expression **XY+X'Z**.
- The term that was eliminated is referred to as **the consensus term**.

3.3 The Consensus Theorem (2) (p70)

- The consensus theorem, given in Eq. (2-18) is

$$XY + X'Z + \boxed{YZ} = XY + X'Z \qquad (3\text{-}20)$$

the consensus terms

- Its Dual form is

$$(X+Y)(X'+Z)\boxed{(Y+Z)} = (X+Y)(X'+Z) \qquad (3\text{-}21)$$

3.3 The Consensus Theorem (3) (p71)

- Example 1:

$$a'b'+ac+bc'+\cancel{b'c}+ab$$

$$= a'b'+ac+bc'+\cancel{ab}$$

$$= a'b'+ac+bc'$$

3.3 The Consensus Theorem (4) (p71)

- Example 2:

$$(a+b+d')(a+b+c')(b+c+d')$$

$$=(a+b+b+d')(a+b+c')(b+d'+c)$$

$$=(\cancel{a+b+b+d'})(a+b+c')(b+d'+c)$$

$$=(a+b+c')(b+d'+c)$$

3.3 The Consensus Theorem (5) (p71)

- Example 3:

$$A'C'D+A'BD+BCD+ABC+ACD'$$

$$=A'C'D+A'BD+BBCD+ABC+ACD'$$

$$=A'C'D+(A'BD+BCBD+ABC)+ACD'$$

$$=A'C'D+(A'BD+\cancel{BCBD}+ABC)+ACD'$$

$$=A'C'D+A'BD+ABC+ACD'$$

3.3 The Consensus Theorem (6) (p71)

- Example 3 in another sense:

$$A'C'D+A'BD+BCD+ABC+ACD'$$

$$=(\mathbf{C'}A'D+\cancel{A'DBD}+\mathbf{C}BD)+ABC+ACD'$$

$$=C'A'D+CB\mathbf{D}+\cancel{ACCB}+AC\mathbf{D'}$$

$$=C'A'D+CBD+ACD'$$

3.3 The Consensus Theorem (7) (p71)

- Example 4:

$$F = ABCD + B'CDE + A'B' + BCE'$$

$$= A\mathbf{B}CD + \mathbf{B'}CDE + A'B' + BCE' + \cancel{ACDCDE}$$

$$= B'CDE + A'B' + (\cancel{ABCCD} + BC\mathbf{E'} + ACD\mathbf{E})$$

$$= (\cancel{B'CDE} + \mathbf{A'}B' + \mathbf{A}CDE) + BCE'$$

$$= A'B' + ACDE + BCE'$$

3.4 Algebraic Simplification of Switching Expressions (1)

(p72)

- 1. **Combining terms (合併代數項):** Use the theorem XY+XY'=X to combine two terms.

$$\boxed{XY + XY' = X}$$

E.g.: abc'd'+abcd'=abd'c' +abd'c=abd'

X Y X Y' X

3.4 Algebraic Simplification of Switching Expressions (2)

(p72)

- 1. **Combining terms (合併代數項):** Use the theorem XY+XY'=X to combine two terms.

$$\boxed{XY + XY' = X}$$

E.g.2: ab'c+abc+a'bc

X Y' X Y X Y X Y'

= (ab'c+abc)+(abc+a'bc)=(acb'+acb)+(bca+bca')

=ac+bc=(a+b)c

3.4 Algebraic Simplification of Switching Expressions (3)

- 1. **Combining terms (合併代數項):** Use the theorem XY+XY'=X to combine two terms. (p72)

$$\boxed{XY + XY' = X}$$

E.g.3: $\underbrace{(a+bc)}_{Y}\ \underbrace{(d+e')}_{X} + \underbrace{a'(b'+c')}_{Y'}\ \underbrace{(d+e')}_{X} = \underbrace{(d+e')}_{X}$

3.4 Algebraic Simplification of Switching Expressions (4)

(p73)

- 2. **Eliminating terms (消去代數項):** Use the theorem X+XY=X to eliminate redundant terms; then try to apply the consensus theorems.

$$\boxed{X+XY=X}$$

$$\boxed{XY+X'Z+YZ=XY+X'Z}$$ (the consensus)

3.4 Algebraic Simplification of Switching Expressions (5)

(p73)

- 2. **Eliminating terms (消去代數項):** Use the theorem X+XY=X to eliminate redundant terms; then try to apply the consensus theorems.

$$\boxed{X+XY=X}$$

$$\boxed{XY+X'Z+YZ=XY+X'Z}$$

E.g.1: $a'b+a'bc=a'b$

X X Y

E.g.2: $a'bc'+bcd+a'b=a'bc'+bcd+a'\mathbf{bd}$

$= a'bc'+bdc+a'bbd = a'bc' + bdc$

Y X' Z X Y Z Y X' Z X

3.4 Algebraic Simplification of Switching Expressions (6)

(p73)

- 2. **Eliminating literals (消去文字字元):** Use the theorem X+X'Y=X+Y to eliminate redundant literals. Simple factoring may be necessary.

$$\boxed{X+X'Y=X+Y}$$

E.g.1: A'B+A'B'C'D'+ABCD'=A'(B+B'C'D')+ABCD'
=A'(B+C'D')+ABCD'=(A'B+ABCD')+A'C'D'
=B(A'+ACD')+A'C'D'=B(A'+CD')+A'C'D'
=A'B+BCD'+A'C'D'

3.4 Algebraic Simplification of Switching Expressions (7)

(p73)

- 2. **Adding redundant terms (加入多餘項):** When possible, the added terms should be chosen so that they will combine with or eliminate other terms.

E.g.1:

$$WX + XY + X'Z' + WY'Z' \qquad (\text{利用重合定理加入} WZ')$$

$$= WX + XY + X'Z' + WY'Z' + WZ' \qquad (\text{消去} WY'Z')$$

$$= WX + XY + X'Z' + WZ' \qquad (\text{消去} WZ')$$

$$= WX + XY + X'Z' \qquad (3\text{-}27)$$

3.4 Algebraic Simplification of Switching Expressions (8)

E.g.2: $\underbrace{A'B'C'D' + A'BC'D'}_{①\,A'C'D'} + A'BD + \underbrace{\cancel{A'BC'D}}_{②} + ABCD + ACD' + B'CD'$

(p73)

$$= A'C'D' + \underbrace{BD(A' + AC)}_{③} + ACD' + B'CD'$$

$$= A'C'D' + A'BD + \underbrace{BCD + ACD'}_{+\,ABC④} + B'CD'$$

$$= A'C'D' + \underbrace{A'BD + \cancel{BCD} + \cancel{ACD'} + \overbrace{B'CD' + ABC}^{\text{重合項}ACD'}}_{\text{重合項}BCD}$$

$$= A'C'D' + A'BD + B'CD' + ABC$$

3.4 Algebraic Simplification of Switching Expressions (9)

(p74)

E.g.3:

A'B'C'+A'B'C

Dual

AC+ACB=AC(1+B)

Dual

$$(\underbrace{A' + B' + C')(A' + B' + C}_{① \ (A' + B')})(B' + C)(A + C)(\cancel{A + B + C})$$

②

$$= \underbrace{(A' + B')(\cancel{B' + C})(A + C)}_{③} = (A' + B')(A + C) \qquad (3\text{-}29)$$

Dual

A'B'+B'C+AC

3.5 Proving Validity of an Equation (1)

(p74)

- 1. Using Truth Table.
- 2. Manipulating the equations until both sides are identical.
- 3. Reduce one side independently until both sides are identical.
- Perform the same operation on both sides of the equations if the operation is reversible.
  - **The compliment is reversible.**
  - **The multiplication is not reversible!!!** (布林代數沒有除法。)

3.5 Proving Validity of an Equation (2) (p74)

- E.g. 1 $A'BD' + BCD + ABC' + AB'D = BC'D' + AD + A'BC$

$$A'BD' + BCD + ABC' + AB'D$$

$$= A'BD' + BCD + ABC' + AB'D + BC'D' + A'BC + ABD$$

（加入 $A'BD'$ 和 ABC' 的重合項）

（加入 $A'BD'$ 和 BCD 的重合項）

（加入 BCD 和 ABC' 的重合項）

$$= AD + A'BD' + BCD + ABC' + BC'D' + A'BC = BC'D' + AD + A'BC \tag{3-30}$$

（刪除 $BC'D'$ 和 AD 的重合項）

（刪除 AD 和 $A'BC$ 的重合項）

（刪除 $BC'D'$ 和 $A'BC$ 的重合項）

3.5 Proving Validity of an Equation (1) (p75)

- E.g. 1 $A'BC'D+(A'+BC)(A+C'D')+BC'D+A'BC'$

$$= ABCD+A'C'D'+ABD+ABCD'+BC'D$$

$$A'BC'D+(A'+BC)(A+C'D')+BC'D+A'BC'$$

（利用吸收定理刪除$A'BC'D$〕

$$=(A'+BC)(A+C'D')+BC'D+A'BC'$$

（利用 (3-3) 式乘開）

$$=ABC+A'C'D'+BC'D+A'BC'$$

（利用重合定理刪除$A'BC'$）

$$=ABC+A'C'D'+BC'D$$

3.5 Proving Validity of an Equation (2) (p75)

- E.g. 1 $A'BC'D + (A' + BC)(A + C'D') + BC'D + A'BC'$

$$= ABCD + A'C'D' + ABD + ABCD' + BC'D$$

Starting from the right:

$$= ABCD + A'C'D' + ABD + ABCD' + BC'D$$

$$= ABC + A'C'D' + ABD + BC'D$$

$$= ABC + \cancel{ABBD} + BC'D + A'C'D'$$

$$= ABC + A'C'D' + BC'D$$

3.5 Proving Validity of an Equation (3) (p76)

- The cancellation law in Boolean Algebra is not valid!

- For add,

$$\boxed{\text{If } x+y = x+z, \text{ then } y=z \text{ ???}}$$

- E.g.,

$$1 + 0 = 1 + 1, \text{ but } 0 \neq 1$$

3.5 Proving Validity of an Equation (4) (p76)

- The cancellation law in Boolean Algebra is not valid!

- For multiplication,

$$\boxed{\text{If } xy = xz, \text{ then } y = z???}$$

- E.g.,

$$0\,y = 0\,z, \text{ but } y \neq z$$

3.5 Proving Validity of an Equation (5) (p76)

- The cancellation law in Boolean Algebra is valid only in one condition.

- If $y = z$,

$$\boxed{x+y = x+z} \qquad \boxed{xy = xz}$$

- When we are to prove an equation is valid, it is NOT permissible to add the same terms to both sides, neither is to multiplicate.

Programmed Exercise 3.1 to 3.6 (pp. 77 to 82)

- Please take exercise by yourself!